David Wadden

Curriculum Vitae

dave.wadden@gmail.com
dwadden.github.io
Google Scholar
Semantic Scholar

Education

2016-2023 **PhD Computer Science**, *University of Washington*, Seattle
Advisor: Hannaneh Hajishirzi
- PhD research focus: Knowledge-intensive NLP for real-world information needs, often motivated by science and health applications.

2006–2010 **BA Physics**, *Amherst College*, Amherst
Phi Beta Kappa, With Distinction

Professional Experience and Internships

2024-Present **Research Scientist**, Google Deepmind, Seattle, WA
Post-training research for Google Gemini.
- Developing RL techniques for long-horizon tasks involving multistep tool use, e.g. SWE and agentic search.

2023-2024 **Research Scientist**, Allen Institute for AI, Seattle, WA
AllenNLP and Semantic Scholar teams.
- Contributed to pretraining / midtraining data creation for the open language model OLMo 2.
- Lead and contributed to research projects focused on (1) AI for scientific literature understanding, and (2) Language model factuality and attribution.

Summer 2021 **Research Intern**, Google, Seattle, WA
Hosts: Nikita Gupta, Kenton Lee, and Kristina Toutanova.

Fall 2019 - Spring 2020 **Research Intern**, Allen Institute for AI, Seattle, WA
Hosts: Kyle Lo and Lucy Lu Wang.

Summer 2018 **Software Engineering Intern**, Google, Kirkland, WA
Host: Evgeny Skvortsov

2011-2014 **Associate Computational Biologist**, The Broad Institute, Cambridge, MA
Supervisor: Aravind Subramanian.

Publications

Conference

2025 Abhilasha Ravichander, Shrusti Ghela, David Wadden, Yejin Choi. **HALoGEN: Fantastic LLM Hallucinations and Where to Find Them.** *ACL 2025*
Outstanding paper.

2025 David Wadden, Kejian Shi, Jacob Morrison, Aakanksha Naik, Shruti Singh, Nitzan Barzilay, Kyle Lo, Tom Hope, Luca Soldaini, Shannon Zejiang Shen, Doug Downey, Hannaneh Hajishirzi, Arman Cohan. **SciRIFF: A Resource to Enhance Language Model Instruction-Following over Scientific Literature**.*EMNLP 2025*

2025 OLMo Team (David Wadden contributed to pretraining data and evaluation). **2 OLMo 2 Furious.** *COLM 2025*

2025 Niklas Muennighoff et al. (David Wadden author #15 of 24). **OLMoE: Open Mixture-of-Experts Language Models.** *ICLR 2025*.

2024 Yanai Elazar*, Jiayao Zhang*, David Wadden*, Bo Zhang, Noah A. Smith. **Estimating the Causal Effect of Early ArXiving on Paper Acceptance.** *CLeaR (Causal Learning and Reasoning) 2024. *Denotes equal contribution*

2024 Fangyuan Xu, Kyle Lo, Luca Soldaini, Bailey Kuehl, Eunsol Choi, David Wadden. **KIWI: A Dataset of Knowledge-Intensive Writing Instructions for Answering Research Questions**. *ACL Findings 2024.*

2024 Jian Guan, Jesse Dodge, David Wadden, Minlie Huang, Hao Peng. **Language Models Hallucinate, but May Excel at Fact Verification.** *NAACL 2024.*

2024 Chao-Chun Hsu, Erin Bransom, Jenna Sparks, Bailey Kuehl, Chenhao Tan, David Wadden, Lucy Lu Wang, Aakanksha Naik. **CHIME: LLM-Assisted Hierarchical Organization of Scientific Studies for Literature Review Support**. *ACL Findings 2024.*

2024 Orion Weller, Kyle Lo, David Wadden, Dawn J Lawrie, Benjamin Van Durme, Arman Cohan, Luca Soldaini. **When do Generative Query and Document Expansions Fail? A Comprehensive Study Across Methods, Retrievers, and Datasets.** *EACL Findings 2024.*

2024 Muhammad Khalifa, David Wadden, Emma Strubell, Honglak Lee, Lu Wang, Iz Beltagy, Hao Peng. **Source-Aware Training Enables Knowledge Attribution in Language Models**. *COLM 2024.*

2023 Yizhong Wang, Hamish Ivison, Pradeep Dasigi, Jack Hessel, Tushar Khot, Khyathi Chandu, David Wadden, Kelsey MacMillan, Noah A. Smith, Iz Beltagy, Hannaneh Hajishirzi. **How Far Can Camels Go? Exploring the State of Instruction Tuning on Open Resources.** *NeurIPS (Datasets and Benchmarks) 2023.*

2023 Ashish Sharma, Kevin Rushton, Inna Wanyin Lin, David Wadden, Khendra G. Lucas, Adam S. Miner, Theresa Nguyen, Tim Althoff. **Cognitive Reframing of Negative Thoughts through Human-Language Model Interaction.** *ACL 2023.*
🏆 **Outstanding paper**.

2022 David Wadden, Nikita Gupta, Kenton Lee, Kristina Toutanova. **Entity-centric query refinement.** *AKBC 2022.*
🏆 **Best Paper Honorable Mention**.

2022 David Wadden, Kyle Lo, Bailey Kuehl, Arman Cohan, Iz Beltagy, Lucy Lu Wang, Hannaneh Hajishirzi. **SciFact-Open: Towards open-domain scientific claim verification.** *EMNLP Findings 2022.*

2022 David Wadden, Kyle Lo, Lucy Lu Wang, Arman Cohan, Iz Beltagy, Hannaneh Hajishirzi. **MultiVerS: Improving scientific claim verification with weak supervision and full-document context.** *NAACL Findings 2022.*

2022 Dustin Wright, David Wadden, Kyle Lo, Bailey Kuehl, Arman Cohan, Isabelle Augenstein, Lucy Lu Wang. **Generating Scientific Claims for Zero-Shot Scientific Fact Checking.** *ACL 2022.*

2021 David Wadden, Tal August, Qisheng Li, Tim Althoff. **The Effect of Moderation on Online Mental Health Conversations.** *ICWSM 2021.*
🏆 **Best Paper for Outstanding Study Design**.

2021 Rahul Nadkarni, David Wadden, Iz Beltagy, Noah A. Smith, Hannaneh Hajishirzi, Tom Hope. **Scientific Language Models for Biomedical Knowledge Base Completion: An Empirical Study.** *AKBC 2021.*

2021 Aida Amini, Tom Hope, David Wadden, Madeleine van Zuylen, Eric Horvitz, Roy Schwartz, Hannaneh Hajishirzi. **Extracting a Knowledge Base of Mechanisms from COVID-19 Papers.** *NAACL 2021.*

2020 David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, Hannaneh Hajishirzi. **Fact or Fiction: Verifying Scientific Claims.** *EMNLP 2020*.

2019 David Wadden, Ulme Wennberg, Yi Luan, Hannaneh Hajishirzi. **Entity, Relation, and Event Extraction with Contextualized Span Representations.** *EMNLP 2019*.

2019 Yi Luan, David Wadden, Luheng He, Hannaneh Hajishirzi, Mari Ostendorf. **A General Framework for Information Extraction using Dynamic Span Graphs.** *NAACL 2019*.

Journal

2026 Akari Asai et al. (David Wadden author # 11 of 25). **Synthesizing scientific literature with retrieval-augmented language models**. *Nature* 650, 857–863.

2019 The ALS Stratification Consortium. **Stratification of amyotrophic lateral sclerosis patients: a crowdsourcing approach**. *Scientific Reports* 9, Article number: 690 (2019).

2018 Oana M Enache, David L Lahr, Ted E Natoli, Lev Litichevskiy, David Wadden, Corey Flynn, Joshua Z Gould, Jacob K Asiedu, Rajiv Narayan, Aravind Subramanian. **The GCTx format and cmap{Py, R, M, J} packages: resources for the optimized storage and integrated traversal of dense matrices of annotated dense matrices**. *Bioinformatics*, bty784.

2017 Aravind Subramanian et al. (David Wadden author #17 of 52). **A Next Generation Connectivity Map: L1000 Platform And The First 1,000,000 Profiles**. *Cell* 171(6):1437-1452.e17.

2017 Ian Smith, Peyton Greenside, Ted Natoli, David L. Lahr, David Wadden, Itay Tirosh, Rajiv Narayan, David E. Root, Todd R. Golub, Aravind Subramanian, John G. Doench. **Evaluation of RNAi and CRISPR technologies by large-scale gene expression profiling in the Connectivity Map**. *PLoS Biol* 15(11):e2003213.

Technical reports

2024 Hamish Ivison, Yizhong Wang, Valentina Pyatkin, Nathan Lambert, Matthew Peters, Pradeep Dasigi, Joel Jang, David Wadden, Noah A. Smith, Iz Beltagy, Hannaneh Hajishirzi. **Camels in a Changing Climate: Enhancing LM Adaptation with Tulu 2**.

Workshop

2021 Neşet Özkan Tan, Niket Tandon, David Wadden, Oyvind Tafjord, Mark Gahegan, Michael Witbrock. **Faithful Reasoning over Scientific Claims.** *AAAI Spring Symposium Series*.

2021 David Wadden, Kyle Lo. **Overview and Insights from the SciVer Shared Task on Scientific Claim Verification.** *Scholarly Document Processing Workshop @ NAACL 2021*.

Honors and Awards

2023 Outstanding Paper, ACL 2023. For *Cognitive Reframing of Negative Thoughts through Human-Language Model Interaction*.

2022 Best Paper Honorable Mention, AKBC 2022. For *Entity-Centric Query Refinement*.

2021 Best Paper Award for Outstanding Study Design, ICWSM 2021. For *The Effect of Moderation on Online Mental Health Conversations*.

2010 William Warren Stifler Prize in Physics, Amherst College.

2007 Bassett Physics Prize, Amherst College.

Open-source software

DyGIE++ (main author): Multitask system for entity, relation, and event extraction (550+ GitHub ★).

SciFact (main author): Dataset and modeling baselines for the SciFact scientific claim verification dataset (200+ GitHub ★).

Open-Instruct (contributor): Open instruction-following language models. (900+ GitHub ★).

Service

Professional service

2021 Organizer, SciVer shared task. Hosted at SDP workshop @ NAACL 2021.

Reviewer: EMNLP, ACL, NAACL, ACL Rolling Review, Journal of Biomedical Informatics.

Departmental service

2019, 2020 PhD admissions committee member.

2020 Prospective PhD student committee member.

Mentoring

Win-Spr '20 Shanchuan Peter Lin, UW CSE MS student → Software engineer, Microsoft.

Win-Spr, '19 Ulme Wennberg, UW CSE MS exchange student → PhD student, KTH Royal Institute of Technology.

Technical skills

Programming languages: Python, JavaScript, R, C, Matlab.

Deep learning frameworks: Tensorflow, Jax, PyTorch, PyTorch Lightning, AllenNLP, Hugging Face Transformers.

Data science packages: Pandas, Numpy, Scikit-Learn, Matplotlib.

Community engagement

2019-2024 **Mentor**, Minds Matter Seattle

Weekly tutoring and college advising for high school students from underrepresented backgrounds.